Individual Homework 05 - Applied

We use the ProPublica COMPAS dataset and the live-coding notebook from the lecture as a starting point.

After importing library packages, loading the data, and training the model, we are ready to create a clean-model fairness baseline.

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CLEAN MODEL BASELINE — Before Any Attack
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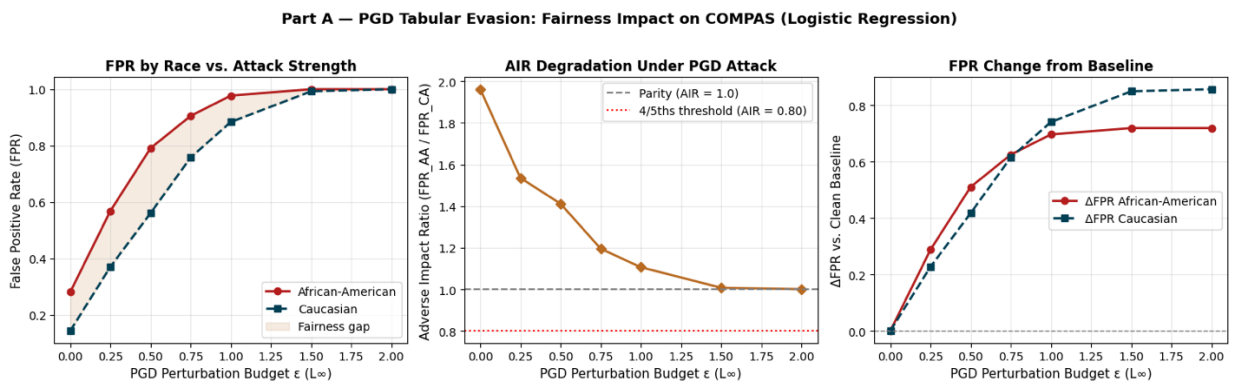
Logistic Regression
  FPR African-American : 0.281
  FPR Caucasian        : 0.143
  AIR (AA / CA)        : 1.961

Gradient Boosted Tree
  FPR African-American : 0.317
  FPR Caucasian        : 0.178
  AIR (AA / CA)        : 1.782

→ Baseline LR AIR stored: 1.961
  We will compare all attack outcomes against these values.
```

Next we run a **Projected Gradient Descent Evasion Audit** on both the linear regression model and gradient-boosted tree model, across the set of epsilon ranging from 0.25 to 2.0. We find that the AIR does not fall below 0.80 within the tested epsilon range:



This Projected Gradient Descent indicates the models are **not equally vulnerable**. In the LR we see larger, more monotone shifts in the FPR and AIR for the same perturbation. In the GBT we see smaller and more irregular changes. This indicates in high-stakes deployments we cannot pick a model based solely on a clean AUC. Model selection must consider robustness and fairness under evasion. A slightly less accurate but more attack-resistant model may be preferable.

Now we move on to a **Label-Flipping Poisoning Attack**. The effect is that the model learns to systematically under-score high-risk individuals from the targeted group. An attack can be considered "stealthy" if the AUC drops ≤ 2pp and the AIR falls below 0.80.

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Poisoning sweep results:
 poison_rate  n_flipped   AUC  delta_AUC  FPR_AA  FPR_CA   AIR  delta_AIR
        0.00          0 0.735      0.000   0.281   0.143 1.961      0.000
        0.02         23 0.733     -0.002   0.270   0.136 1.986      0.024
        0.05         57 0.734     -0.001   0.240   0.123 1.948     -0.014
        0.08         92 0.732     -0.002   0.213   0.116 1.840     -0.122
        0.10        115 0.735      0.000   0.196   0.101 1.931     -0.030
        0.15        172 0.732     -0.003   0.153   0.072 2.134      0.173
        0.20        230 0.732     -0.002   0.130   0.054 2.399      0.438
        0.25        287 0.730     -0.004   0.112   0.042 2.677      0.715
        0.30        345 0.731     -0.003   0.097   0.032 3.010      1.049
```
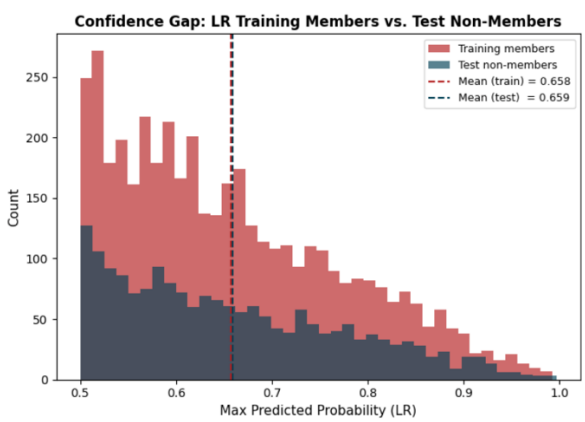
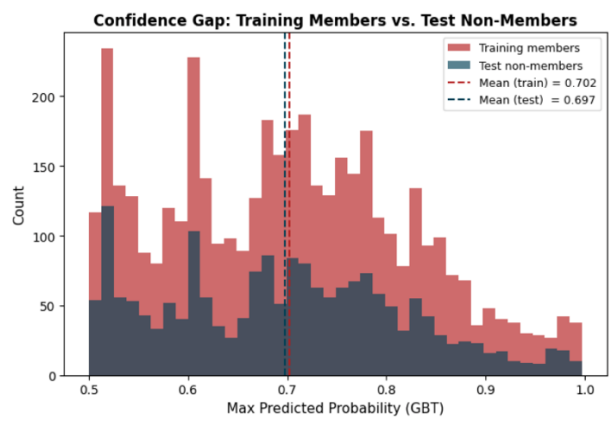

No stealth threshold was found in the tested range.

A **Population Stability Index-based drift monitor** will almost certainly not detect either attack. Label-flip poisoning only changes the labels for a subset of records. Since the feature distribution remains essentially the same, the PSI on each feature stays near zero. We need explicit label- and outcome-based fairness monitoring, such as AIR and FPR gaps, in addition to monitoring for feature drift.

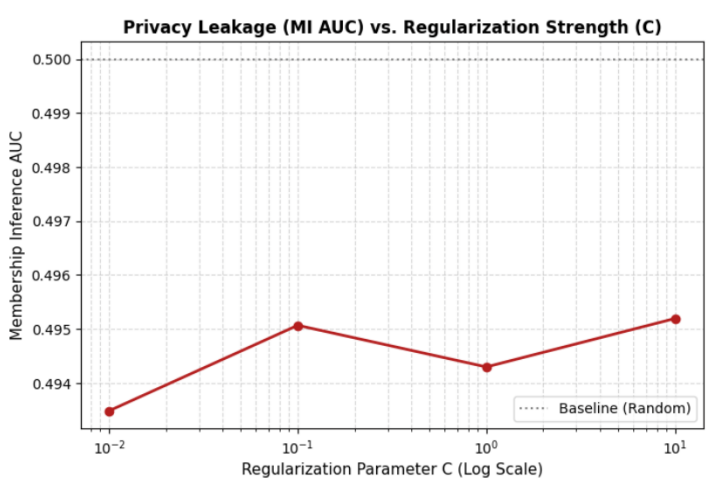
We move on to **Membership Inference** and compute a shadow-model MI AUC for both models. It is helpful to visualize the confidence gaps:





For both models, the histograms nearly completely overlap and the confidence gaps are very tiny. This indicates both models have no meaningful privacy leakage or exploitable overfitting signal. Generalization is smooth and membership inference attacks fail.

For our last step we add L2 regularization to the linear regression model.



The curve stays between 03494 and 0.5 for every tested value of c. This indicates the linear regression model never becomes more vulnerable to membership inference, even when regularization is weakened or strengthened. The practical trade-off is privacy vs. utility, but because the MI AUC is so consistent, generalization is smooth and privacy leakage is minimal, so regularization does not increase privacy or cost us accuracy.

Reflection:

Across all three analyses, the **highest-risk finding** comes from the label-flipping poisoning attack. Unlike the PGD evasion audit (which required large perturbations to meaningfully shift error rates) and the membership-inference tests (which showed essentially no privacy leakage) the poisoning attack was able to *severely* distort group-level fairness while accuracy was almost unchanged. In the results, the AIR for the logistic regression model rose from 1.96 to over 3.0, even though the test AUC rose by less than 0.5 percent. This means an attacker could push the model to treat racial groups very differently, without triggering standard performance alarms!

A **proactive mitigation** would be to build fairness checks directly into the training of the model. For example, monitoring AIR or FPR gaps during retraining would flag the poisoning patterns we observed. Even at a 5-10% poison rate, AIR shifted by 0.12-0.17, which is beyond acceptable levels. Rejecting or reweighting data when these fairness metrics drift would prevent the model from learning the poisoned pattern in the first place.

A **reactive mitigation** is to deploy continuous fairness monitoring. Instead of watching only accuracy or feature drift, the system should track AIR and FPR by race over time. This would have caught the attack as soon as AIR moved from 1.96 to 1.84 at an 8% poison rate. The tradeoff is that this monitoring may trigger more frequent interventions for one group than another, but that imbalance reflects the underlying harm: this attack disproportionately reduces accuracy for African-American defendants while falsely improving it for Caucasian defendants.

Together, these results show that the **most serious vulnerability** is not evasion or privacy leakage, but quiet fairness degradation. The mitigations that work best are the ones that keep fairness metrics visible and enforceable throughout the model's lifecycle.